Export-Induced Spatial Divergence

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March 13, 2026

Abstract

We study how export market access shapes within-industry spatial concentration, exploiting cross-industry variation in trade policy uncertainty eliminated by China's WTO accession in 2001. Using geolocated firm data, we measure employment density in concentric rings around pre-existing cluster centers. Only the initially largest clusters grow: a one-standard-deviation larger reduction in trade policy uncertainty raises employment density by 21.2% at 10–20km from the cluster center for an average top-quartile cluster by 2007, with effects decaying to zero beyond 100km. This Matthew effect is driven entirely by new firm entry, private and foreign, not by incumbent expansion. Export-exposed clusters also generate positive density spillovers to upstream and downstream industries but crowd out industries competing for the same workers.

Topics: firm location, spatial concentration, trade openness

JEL Codes: F6, F14, R12

First circulated as "Trade Liberalization, Agglomeration, and Urbanization: Evidence from China". We are very grateful to Jan Bakker, Loren Brandt, David Dorn, Carsten Eckel, Harald Fadinger, Lisandra Flach, Inga Heiland, Ines Helm, Martina Magli, Isabela Manelici, Andreas Moxnes, Ralph Ossa, Claudia Steinwender, Karen Helene Ulltveit-Moe, Nico Voigtländer, Horng Chern Wong, Yuan Zi, and workshop participants at the 2022 Göttingen Trade Workshop, the 2022 Danish International Economics Workshop, the 2023 Workshop in International Economic Networks (WIEN), the 2023 UEA European Meeting, and other seminar participants for their valuable comments and suggestions. Support from the Deutsche Forschungsgemeinschaft through CRC TRR 190 (project number 280092119) and CRC TR 224 (Project A03 and Project B06) are gratefully acknowledged. All mistakes are our own.

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1. Introduction

The unequal distribution of production across space is one of the most prevalent features of modern economies (Marshall 1890; Duranton and Overman 2005; Combes et al. 2012; Redding 2020). A large literature has studied how international trade shapes this distribution, but causal evidence centers predominantly on the market access channel, through which reduced trade costs raise returns in locations with better access to foreign markets, concentrating activity near ports and borders (Redding and Sturm 2008; Redding and Rossi-Hansberg 2017; Fajgelbaum and Redding 2022; Coşar and Fajgelbaum 2016). Regional comparative advantage and location fundamentals offer a distinct channel through which trade shapes the spatial distribution of economic activity. Regions with stronger pre-existing industry clusters are better positioned to exploit new export opportunities, so trade liberalization amplifies existing spatial inequalities rather than dissolving them (Bakker et al. 2024). We explore this channel, asking how export market opening shapes the spatial distribution of manufacturing activity within China following WTO accession.

We document a pronounced Matthew effect in which export market opening benefits only the initially largest industry clusters. Among pre-existing clusters ranked by 1998 employment, only those in the top quartile grow following WTO accession; clusters in the bottom three quartiles are unaffected. Within top-quartile clusters, employment density rises most sharply 10–20km from the center (21.2% per standard-deviation increase in the NTR gap) and attenuates to zero beyond 100km. This growth is driven entirely by new firm entry; incumbent employment is unchanged. Private and foreign entrants account for the expansion, while state-owned and collectively owned enterprises contract. Export-exposed clusters generate positive co-agglomeration spillovers to upstream and downstream industries but crowd out industries competing for the same workers, reflecting imperfect local labor market integration. China's WTO accession amplified pre-existing spatial inequalities rather than diffusing them.

We identify the causal effect of export market opening using the NTR gap (Pierce and Schott 2016; Handley and Limão 2017), which measures the reduction in trade policy uncertainty when China joined the WTO in 2001. Prior to accession, China's preferential US tariff rates required annual renewal by Congress and could be revoked, so industries facing a larger NTR gap operated under greater uncertainty about future market access. Because the gap was determined by US domestic political economy considerations, it is plausibly exogenous to Chinese firm location decisions. We interact the NTR gap with each industry-prefecture cluster's initial size, measured by 1998 employment quartile, and track employment density in geolocated concentric rings around each cluster center using the Annual Survey of Industrial Firms.

The spatial structure of the response is itself informative. Growth concentrates within 100km of the cluster center, consistent with agglomeration externalities operating at the scale of commuting zones and regional labor markets. The attenuation in effects at the 0–10km core ring and their peak at 10–20km is consistent with land-market crowding at the cluster center. Beyond 100km, the effects are zero, indicating that trade-driven agglomeration does not propagate indefinitely across space.

The entry-driven margin deserves emphasis. We decompose the cumulative employment change by 2007 into five components: incumbent firm employment growth, incumbent firm exit, new firm entry, industry in-switching, and industry out-switching. The 4th quartile effect is accounted for almost entirely by new entrants. The ownership decomposition reinforces this interpretation: private and foreign firms are the entrants driving the 4th quartile effect, while state-owned and collectively owned enterprises are contracting. The divergence is a market-driven phenomenon, not a product of state resource allocation.

Trade liberalization also affects industries that are not directly exposed to the export shock. Upstream and downstream industries co-agglomerate with expanding export sectors, but only in 4th quartile clusters, where supply-chain neighbors grow when the exposed industry grows, conditional on initial cluster size. Industries competing for the same workers, by contrast, lose employment density in 4th quartile clusters when an exposed industry expands. This negative labor-pooling spillover implies that local labor markets are not well integrated across industries or regions, so rising labor demand in export clusters crowds out nearby competitors rather than drawing workers in from outside. A Spatial-Melitz model in which cities inherit productivity distributions from historical size rationalizes these patterns. Tariff reductions tighten selection, and the gains from that tightening concentrate in cities with thicker upper tails of the productivity distribution, which are the initially largest.

Our most direct contribution is to the literature on trade and agglomeration. Bakker et al. (2024) embed Melitz-style heterogeneous firms in a quantitative spatial model and predict that trade liberalization shifts workers toward larger cities with denser exporter populations. Helm (2020) documents that export shocks increase spatial concentration in Germany through vertical linkages. Quantitative spatial models predict that liberalization concentrates activity near ports or borders (Redding and Sturm 2008; Redding and Rossi-Hansberg 2017; Fajgelbaum and Redding 2022); Coşar and Fajgelbaum (2016) and Nagy (2022) show concentration in large cities or designated trading places. We contribute three things to this literature. First, we provide direct causal evidence from a developing country using geolocated firm records, avoiding the administrative-boundary distortions that affect city- and region-level analyses.[1] Second, we recover the full distance gradient of agglomeration effects, showing the reach of trade-driven concentration is bounded at roughly 100km. Third, we decompose the spatial response into firm-level margins, establishing entry rather than incumbent expansion as the operative mechanism.

We also contribute to the literature on firm sorting and agglomeration. Gaubert (2018) shows that productive firms sort into large cities, amplifying spatial productivity and wage differences. We show that firm sorting and export market access intensify each other. Initial sorting implies that large places benefit more from trade liberalization, which then further induces entry of new firms into already-large clusters.

[1] Notable exceptions that do use firm-level geocode data are Alfaro and Chen (2014) and Alfaro et al. (2019).

On co-agglomeration, Ellison et al. (2010) and Combes et al. (2012) document patterns consistent with input-output, labor-pooling, and technology channels. Curuk and Vannoorenberghe (2017) and Helm (2020) show that trade-induced growth in one industry spills positively to vertically linked industries. We confirm these positive supply-chain spillovers but also document negative labor-pooling spillovers, a finding that highlights the role of imperfect labor mobility in limiting the spatial diffusion of trade gains. This complements the literature on local labor market effects of China's WTO accession (Autor et al. 2013; Acemoglu et al. 2016). We show that on the Chinese side, the benefits of accession were concentrated in pre-existing large clusters.

The remainder of the paper is organized as follows. Section 2 describes the data and variable construction. Section 3 presents the empirical strategy. Section 4 presents the main results on spatial divergence. Section 5 reports robustness checks. Section 6 examines co-agglomeration with upstream, downstream, and labor-pooling industries. Section 7 develops the theoretical framework. Section 8 concludes.

2. Data And Variable Construction

2.1. Data Sources And Sample

The primary data source is the Annual Survey of Industrial Firms (ASIF), conducted by China's National Bureau of Statistics (NBS) and covering the years 1998–2007. The survey encompasses all state-owned enterprises and private firms with annual sales of at least 5 million RMB (approximately $770,000 at the prevailing exchange rate). Subsidiaries of larger firms enter the sample independently as long as they constitute separate legal units (Brandt et al. 2014). In 1998, 88.9% of firms reported a single production plant, a share rising to 96.6% by 2007, so ASIF approximates an establishment-level survey (Brandt et al. 2014). Using the 2004 economic census, Brandt et al. (2012) document that while ASIF accounts for only 20% of all industrial firms, these firms employ 71.2% of the industrial workforce, produce 90.7% of output, and generate 97.5% of exports. Though a unique firm identifier allows us to track firms over time, some identifiers change due to mergers, acquisitions, restructuring, or administrative reassignment. We follow the five-step matching procedure of Brandt et al. (2012) to obtain a consistent panel[2].

We impose several sample restrictions. We drop firms with missing employment, exports, or wage bill, and firms reporting negative exports or negative gross output. Following Brandt et al. (2017), we exclude firms with fewer than eight employees, as these fall under a different legal regime. To remove pure re-exporters, we drop firms with an export-to-output ratio exceeding 2. The Chinese Industrial Classification (CIC) changed between 2002 and 2003; we apply the crosswalk from Brandt et al. (2012) to maintain a consistent industry classification and remove industries reclassified as services or non-manufacturing.[3] We also drop other non-manufacturing industrial activities (e.g., mining, oil refining). Finally, we drop firms whose addresses could not be resolved to geographic coordinates via Baidu Map API v3.0; coordinates are converted from GCJ02 to WGS84. Appendix Table A2 reports the geocoding confidence distribution: over 99% of geocoded observations carry a confidence score implying a location error of at most 100 meters. These restrictions yield 142,393 firms in 1998 and 310,064 firms in 2007. Aggregate trends in the ASIF sample are reported in Appendix Table A1.

Data on NTR gaps are from Pierce and Schott (2016), matched to the Chinese Industry Classification using the product-industry concordance of Brandt et al. (2012) and HS 6-digit concordances from the World Integrated Trade Solution (WITS). For robustness controls, we supplement the ASIF data with prefecture-level information on road intensity, high-speed railway connections, and special economic zones from Martin and Zhang (2021).

[2]The process involves matching first on the unique firm identifier, then on firm name, then on the combination of the legal representative's name, prefecture code, and industry code, and finally on the combination of phone number and prefecture code, and on founding year, 6-digit region code, industry code, address, and primary product name.

[3]Affected CIC codes: 1711, 1712, 1713, 1714, 2220, 3648, 3783, 4183, and 4280.

2.2. Measuring Spatial Concentration

Standard measures of spatial concentration (employment or output aggregated by administrative unit) conflate the geographic extent of industry clusters with the scale of administrative geography. A cluster straddling a prefecture boundary appears as two distinct units; two prefectures with the same total employment look identical regardless of whether that employment is tightly packed or dispersed. These distortions preclude comparison across regions and make it impossible to ask how far from a cluster center the effects of trade liberalization extend.

We address this with a concentric ring approach. For each industry j and prefecture p, we define the cluster center as the employment-weighted centroid of all firms in 1998. We then construct concentric rings at 10km intervals, from 0–10km up to 140–150km, around that center and measure employment density within each ring. Rings from different industry-prefecture pairs overlap freely, so the measure is unconstrained by administrative borders.

Formally, spatial concentration for industry j, prefecture p, ring r, and year t is

$$concentration^t_{jpr} = \frac{L^t_{jpr}}{Area_{pr}} = \frac{\sum_f L^t_{jprf}}{Area_{pr}}, \tag{1}$$

where $L^t_{jpr} = \sum_f L^t_{jprf}$ is total employment across all firms f in ring r, $Area_{pr}$ is ring area in units of 10,000m^2, and j, p, r index industry, prefecture, and ring. The data span 161 three-digit industries and 336 prefectures, yielding rings up to 150km from each cluster center. Figure 1 illustrates the construction for Basic and Raw Chemicals in Shanghai in 1998: blue dots mark individual firms (scaled by employment), the red dot marks the employment-weighted cluster center, and the concentric circles define the ring boundaries.

A central question in our analysis is whether trade liberalization affects clusters of different initial sizes differently. To capture this heterogeneity, we rank all industry-prefecture pairs within each industry by total employment in 1998 and assign them to quartiles Q1 (smallest) through Q4 (largest). These assignments are fixed at their 1998 values and serve

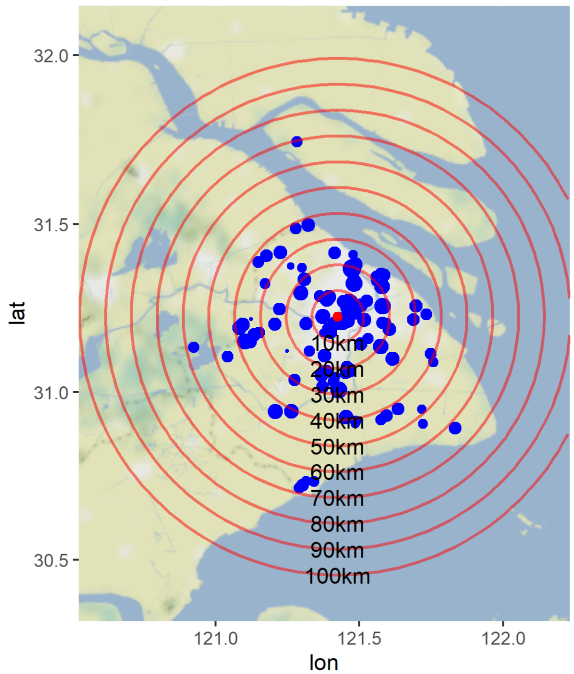


Figure 1: Concentric rings for Basic and Raw Chemicals in Shanghai, 1998. Blue dots are individual firms (scaled by employment). The red dot is the employment-weighted cluster center.

as the key cross-sectional heterogeneity dimension throughout the paper. Figure 2 motivates this focus for Agricultural Machinery Manufacturing: the clusters that were largest in 1998 are precisely those that expanded by 2007.

Table 1 reports summary statistics on the ring-level data. Employment and firm density decline with distance from the cluster center and increase sharply with quartile rank: average employment density at 10–20km is 208 workers per 10,000m^2 for Q4 clusters, compared with 9 for Q1 clusters. The exporter share also rises with quartile rank at every ring. This gradient is consistent with the joint sorting of productive firms into large agglomerations, as Gaubert (2018) and Combes et al. (2012) predict. Bakker (2021) and Bakker et al. (2024) document a similar pattern in France and the 2004 Chinese industrial census, though without distinguishing across industries.

Table 1: Summary of concentric ring data

	Q1: Mean	SD	Q2: Mean	SD	Q3: Mean	SD	Q4: Mean	SD
Employment Density								
0-10km	45.148	189.812	88.801	249.438	177.974	379.525	737.725	2015.619
10-20km	8.857	68.312	18.658	80.476	42.281	133.949	208.478	570.903
20-30km	6.724	39.344	11.857	40.967	26.734	120.222	110.985	378.095
30-40km	6.776	33.662	11.127	45.863	21.08	81.539	72.317	223.431
40-50km	8.308	39.942	12.474	64.438	19.38	68.061	59.735	192.563
50-150km	8.978	43.278	13.028	51.608	20.456	75.806	45.679	151.498
Firm Density								
0-10km	.273	.407	.323	.538	.496	.886	1.557	3.807
10-20km	.047	.155	.089	.25	.176	.428	.727	1.678
20-30km	.034	.116	.058	.147	.114	.32	.411	1.038
30-40km	.032	.098	.048	.137	.09	.261	.283	.721
40-50km	.034	.107	.05	.167	.085	.247	.225	.562
50-150km	.037	.12	.051	.151	.083	.23	.172	.447
Exporter Share								
0-10km	.133	.321	.208	.373	.26	.379	.319	.358
10-20km	.177	.348	.2	.355	.234	.358	.278	.334
20-30km	.17	.332	.185	.343	.222	.348	.271	.336
30-40km	.181	.337	.182	.33	.215	.338	.261	.33
40-50km	.184	.327	.188	.327	.212	.332	.256	.327
50-150km	.183	.314	.192	.315	.212	.315	.247	.314
Change Employment Density								
0-10km	-.339	195.727	-27.802	216.668	-72.215	347.092	-232.278	1625.439
10-20km	5.291	67.781	4.238	75.304	5.805	114.401	32.505	430.404
20-30km	2.198	33.882	1.589	41.528	5.84	113.886	34.28	307.653
30-40km	2.194	29.223	1.88	39.206	5.51	61.783	20.499	169.712
40-50km	.559	36.864	1.545	51.448	3.514	53.539	16.922	129.119
50-150km	.912	44.37	1.564	38.861	2.953	61.211	11.02	113.177

Notes: The unit of observation is an industry-prefecture-ring cell. Rings are defined as concentric bands at 10km intervals around the 1998 employment-weighted cluster centroid. Quartiles Q1–Q4 rank industry-prefecture clusters by total employment in 1998, within each industry; Q4 is the largest. Employment density is total employment divided by ring area, measured in workers per 10,000m^2. Firm density is the number of firms per 10,000m^2. Exporter share is the fraction of firms reporting positive exports. Change employment density is the 1998–2007 change in employment density in the same units. Data are from the Annual Survey of Industrial Firms (ASIF), 1998 and 2007.

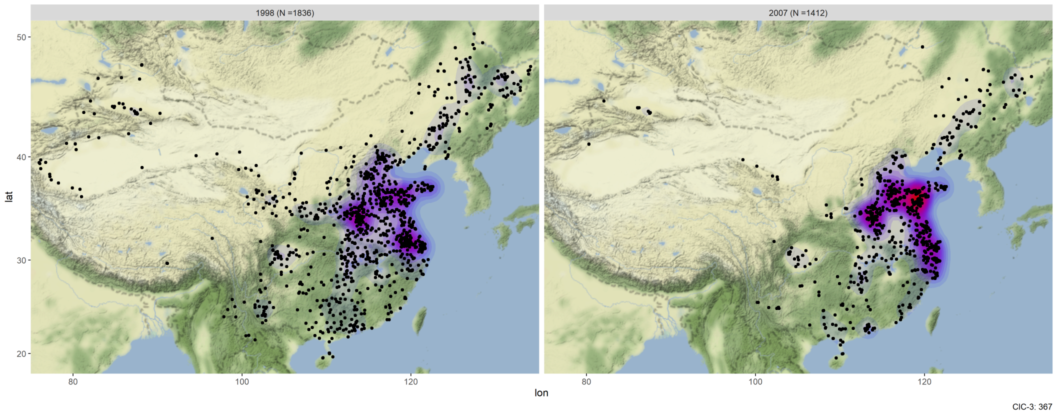


Figure 2: Employment density of Agricultural Machinery Manufacturing, 1998 (left) and 2007 (right). The clusters that were largest in 1998 grew the most.

2.3. Trade Policy Variable: The NTR Gap

Identification exploits cross-industry variation in trade policy uncertainty. Before China's WTO accession in December 2001, the Jackson–Vanik amendment necessitated annual congressional approval for the maintenance of Normal Trade Relations (NTR) status. Although China received NTR status in 1980, the renewal process subjected exporters to the constant risk of high non-NTR tariffs. WTO accession permanently removed this requirement and the associated policy risk.

The NTR gap, defined as the difference between non-NTR and NTR tariffs (Pierce and Schott 2016), quantifies the trade policy uncertainty eliminated by WTO accession. Since the US Congress determined these gaps in the 1930s based on domestic political considerations rather than Chinese industrial characteristics or manufacturing geography, the measure is treated as exogenous to Chinese firm location decisions. Identification relies on the interaction of this cross-industry variation with the timing of WTO accession to estimate the causal impact of trade liberalization on spatial concentration.

The mean NTR gap is 30.5 percentage points with a standard deviation of 14.0 percentage points; the variable is fixed across regions and does not vary over time. Table 2 reports the distribution of the NTR gap alongside the initial employment of each quartile.

Table 2: Summary of NTR-Gap and initial cluster size

	Mean	SD	Min	Max
NTR Gap	30.52	14.04	0	71.68
Initial Size				
Q1	164.05	201.91	8	2272
Q2	620.3	611.44	50	6104
Q3	1612.14	1540.46	58	14128
Q4	7060.33	10312.2	91	165953

Note: Initial size denotes the number of workers in a specific industry-prefecture pair in 1998.

3. Empirical Strategy

3.1. Identification Strategy

Our identifying variation combines three elements. First, the NTR gap varies across industries: industries with a higher gap faced greater trade policy uncertainty before WTO accession. Second, China's WTO accession in December 2001 eliminated this uncertainty in a nationwide shock common to all industries. Third, within each industry, clusters in different prefectures differ in initial size, which may condition how strongly they respond to the reduction in uncertainty.

We interact these three elements to estimate the causal effect of trade liberalization on spatial concentration. The strategy compares how employment density evolves differently across size quartiles as a function of the industry-level NTR gap. The key identifying assumption is that, absent WTO accession, employment density in highly exposed Q4 clusters would have evolved in parallel to employment density in smaller clusters, conditional on province and sector fixed effects and the controls described below. We validate this assumption using pre-2001 coefficients in Section 4.

3.2. Main Specification

Following Dix-Carneiro and Kovak (2017), we take changes in employment density relative to 1998 and estimate the model separately for each ring r and each year $t \in \{1999, \ldots, 2007\}$:

$$Y^t_{prj} - Y^{1998}_{prj} = \sum_{q=1}^{4} \beta_{q,t}\, \mathrm{NTR}_j \times Q_{pjq} + X^t_{prj} + \mu_P + \gamma_s + \epsilon^t_{prj}, \tag{2}$$

where j indexes three-digit industries, p prefectures, and r concentric rings. The dependent variable $Y^t_{prj} - Y^{1998}_{prj}$ is the change in employment density (equation 1) relative to 1998, measured in workers per 10,000m^2. NTR_j is the industry NTR gap, fixed at its 1998 value. Q_{pjq} is a dummy equal to one if the cluster of industry j in prefecture p belongs to initial size quartile q. The coefficient $\beta_{q,t}$ measures the cumulative effect of a unit increase in the NTR gap for quartile q by year t.

The control vector X^t_{prj} includes changes in Chinese input and output tariffs between t and 1998, to absorb the independent effect of domestic import liberalization; initial shares of foreign- and private-owned capital in 1998, capturing pre-existing industry composition and local policy orientation; and the share of firms in special economic zones, using data from Martin and Zhang (2021). Province fixed effects μ_P and two-digit sector fixed effects γ_s absorb time-invariant regional and broad-industry confounders. Standard errors are clustered at the three-digit industry level.

The identification strategy relies on the parallel-trends assumption, which the event-study framework in equation 2 validates. Pre-accession coefficients are statistically indistinguishable from zero, whereas the post-2001 divergence is immediate and pronounced. To isolate the reduction in trade policy uncertainty from contemporaneous shocks, the analysis incorporates controls for several potential confounders. Furthermore, the results reported in Section 4 are robust to the inclusion of prefecture-level infrastructure developments, such as road expansion and high-speed rail connectivity, and aggregate prefecture-level trade exposure measured by the employment-weighted average NTR gap. These tests collectively confirm that the observed shifts in employment density reflect the WTO accession shock rather than pre-existing trends or omitted geographic and policy variables.

3.3. Pooled Specification

Estimating equation 2 year by year restricts the available fixed effects and, in particular, precludes prefecture-by-year fixed effects. If Q4 clusters are concentrated in larger, faster-growing prefectures, not controlling for prefecture-specific time trends could conflate a differential growth path with a response to trade liberalization. To address this, we also estimate:

$$Y_{prjt} - Y_{prj}^{1998} = \sum_{q=1}^{4} \beta_q \, \mathrm{NTR}_j \times Q_{pjq} \times \mathrm{Post}_t + X_{prjt} + \mu_{pt} + \gamma_j + \epsilon_{prjt}, \tag{3}$$

where Post_t equals one for $t > 2001$. The prefecture-by-year fixed effects μ_{pt} absorb all prefecture-specific trends, including infrastructure investment and the direct effect of aggregate trade exposure. Three-digit industry fixed effects γ_j absorb time-invariant industry characteristics. X_{prjt} includes changes in Chinese input and output tariffs, initial ownership shares, and SEZ shares; the additional robustness controls for road length, high-speed rail, and employment-weighted prefecture NTR gap are absorbed by μ_{pt} and need not be included separately. We report results from this specification in Section 5.

4. Spatial Divergence: Main Results

4.1. Between-Cluster Divergence

The central prediction of combining Melitz (2003) with Gaubert (2018), as formalized in Bakker (2021), is that trade liberalization disproportionately benefits large agglomerations. Sorting and agglomeration externalities concentrate productive, export-active firms in large clusters; it is these firms that gain the most from reduced trade policy uncertainty. Figure 3 tests this prediction directly. It plots the estimated coefficients $\hat{\beta}_{q,t}$ from equation 2 for each quartile and year, using total employment density within 100km of the cluster center as the dependent variable.

Before 2001, no quartile shows a statistically significant trend. The coefficients for 1999 and 2000 are indistinguishable from zero for all four quartiles, validating the parallel-trends assumption and confirming that industries with larger NTR gaps were not on differential growth paths prior to WTO accession. After 2001, a sharp divergence emerges: Q4 clusters experience a large and growing increase in employment density, while Q1, Q2, and Q3 clusters show no significant response. The divergence is absent before the accession date and appears immediately afterward, consistent with a causal interpretation.

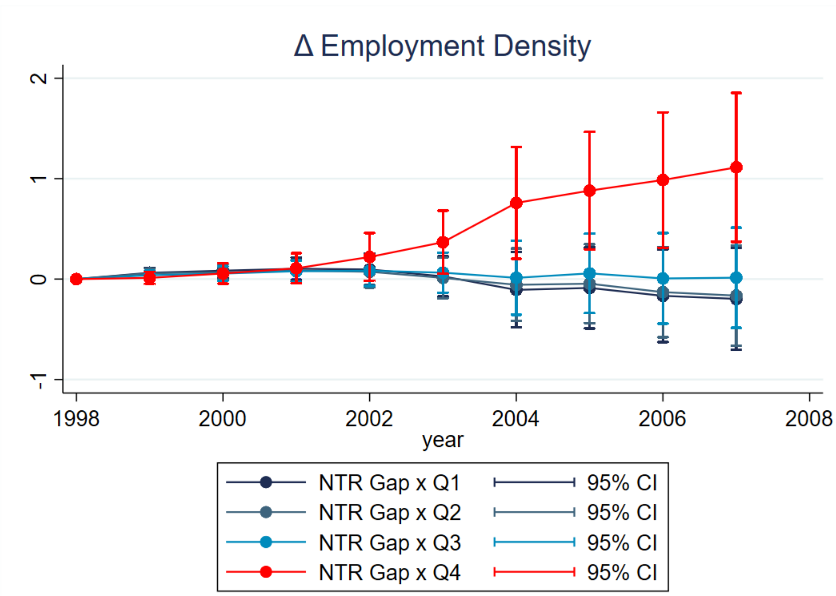


Figure 3: Change in employment density within 100km of the cluster center, by initial size quartile and year. Plotted coefficients are $\hat{\beta}_{q,t}$ from equation 2, estimated separately for each year. The dependent variable is the change in employment density relative to 1998. The vertical dashed line marks China's WTO accession in December 2001. Standard errors are clustered at the three-digit industry level; 95% confidence intervals shown.

The central result is a Matthew effect, whereby initially large clusters grow larger while smaller clusters stagnate. Because sorting concentrates the most productive and export-oriented firms in Q4 clusters, these clusters host firms which benefit the most from the reduction in effective trade costs due to WTO accession. Smaller clusters, with lower concentrations of exporting firms, benefit little from the same reduction in trade policy uncertainty.

4.2. The Distance Gradient

A distinctive feature of our design is the ability to measure how far the agglomeration effect extends from the cluster center. Figure 4 contrasts the event-study coefficients at two distances: 10–20km (top) and 50–60km (bottom). At 10–20km, the Q4 pattern mirrors the 100km result closely: no pre-2001 trend, then a sharp and growing increase after WTO accession. At 50–60km, the Q4 coefficient is smaller and noisier, indicating that the same trade shock generates a weaker spatial footprint at greater distances.

Figure 5 plots the full distance profile by displaying the cumulative Q4 effect by 2007 ($\hat{\beta}_{4,2007}$) for each ring. The profile peaks at 10–20km, declines steadily with distance, and becomes statistically indistinguishable from zero beyond 100km.[4] The magnitude of the effect is substantial. A one-standard-deviation larger NTR gap (14 percentage points) is associated with a 21.2% increase in employment density at 10–20km for an average Q4 cluster by 2007.

4.3. Decomposition By Firm Margin

The aggregate density patterns in Figures 3–5 do not identify which firm-level adjustment margins drive the Q4 expansion. One possibility is that a pure Melitz (2003) mechanism, without any agglomeration externality, accounts for the result: Q4 clusters may simply contain more productive firms, which scale up the most after liberalization. In this case, the positive Q4 effect would reflect within-firm employment growth by incumbent exporters, not the sorting of new entrants into large agglomerations.

We decompose the cumulative change in employment density by 2007 into five mutually

[4] We exclude the 0–10km ring throughout. Cluster centers are defined as employment-weighted centroids in 1998; because the innermost ring covers the smallest area, even small differences in geocoding precision produce large variance in density. The resulting estimates are too imprecise to be informative. Furthermore, as Baum-Snow et al. (2017) show, industrial activity moved out of the immediate city center in China during this time, driven by radial transport networks.

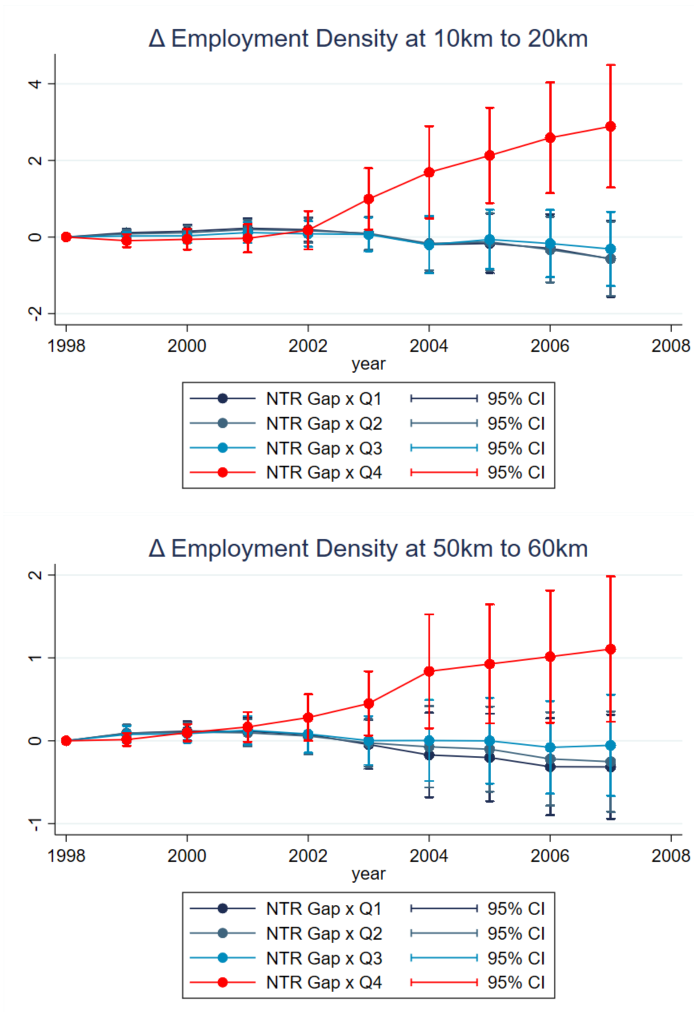


Figure 4: Change in employment density at 10–20km (top) and 50–60km (bottom) from the cluster center, by initial size quartile and year. Plotted coefficients are $\hat{\beta}_{q,t}$ from equation 2, estimated separately for each ring–year combination. The vertical dashed line marks WTO accession in December 2001. Standard errors clustered at the three-digit industry level; 95% confidence intervals shown.

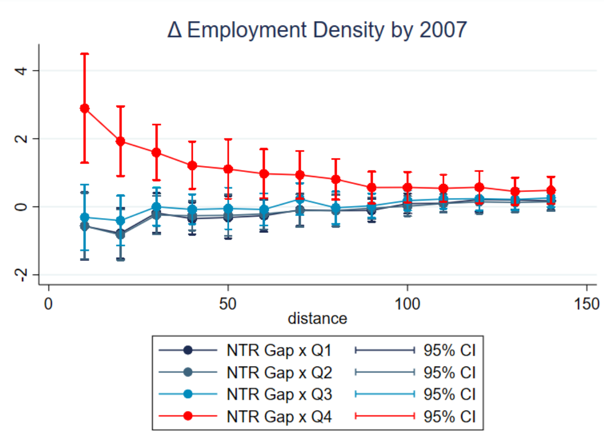


Figure 5: Cumulative change in employment density by 2007, by ring and initial size quartile. Plotted coefficients are $\hat{\beta}_{q,2007}$ from equation 2, estimated separately for each ring. Each point on the horizontal axis corresponds to the midpoint of a 10km ring (e.g., 15km represents the 10–20km ring). Standard errors clustered at the three-digit industry level; 95% confidence intervals shown. The 0–10km ring is excluded.

exclusive margins:

1. *Incumbents*: change in employment among firms present in the same industry-prefecture cell in both 1998 and 2007;
2. *Exits*: employment loss from firms active in 1998 that had left the sample by 2007;
3. *Entry*: employment of firms first observed in 2007 that were not present in 1998;
4. *Industry in-switching*: employment of firms that operated in a different industry in 1998 and switched into the liberalizing industry by 2007;
5. *Industry out-switching*: employment lost from firms that operated in the industry in 1998 but switched to a different industry by 2007.

This decomposition is exact up to firm relocation across rings, which is rare and contributes negligibly to the observed patterns.

Figure 6 presents the cumulative 2007 Q4 coefficient for each margin across rings. The positive aggregate effect in Q4 is driven almost entirely by new firm entry. The entry panel reproduces the distance profile of Figure 5, peaking at 10–20km and attenuating with distance.

Incumbents, by contrast, contribute negatively or near-zero: surviving pre-liberalization firms do not expand in large clusters and may even contract. Exits add a further small negative contribution. The net effect of in-switching is positive, as more incumbent firms switch into the liberalizing industry in Q4 clusters than leave it, while out-switching is a small offsetting drag.

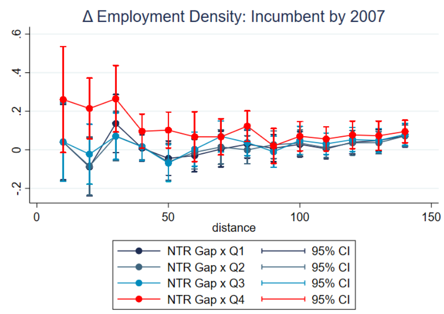
The Q4 spatial divergence reflects the extensive margin of agglomeration. New firms enter preferentially in large clusters, attracted by the thick upper productivity tail, deep labor pools, and supply-chain networks already concentrated there. The concurrent increase in competition from entrants may also squeeze surviving incumbents, contributing to the negative incumbent coefficient.

4.4. Ownership Heterogeneity

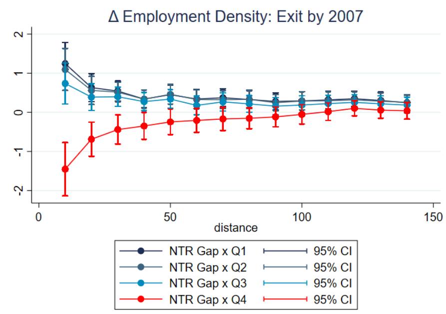
China's post-WTO period coincided with a dramatic expansion of the private sector and an influx of foreign direct investment, alongside the continued operation of state-owned and collectively owned enterprises. Figure 7 decomposes the aggregate density effect by ownership type: private domestic firms, foreign-owned enterprises, and state- and collectively owned enterprises.

Private domestic and foreign-owned firms account for all of the Q4 density increase. Both ownership panels replicate the spatial profile, peaking at 10–20km and decaying beyond 100km. State- and collectively owned enterprises, by contrast, contract in Q4 clusters: their coefficient is negative across the distance profile. In initially smaller clusters, SOEs and collective firms show a modest positive response.

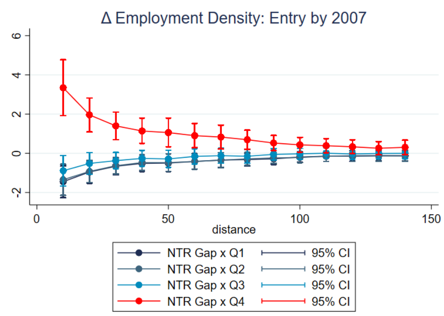
The ownership pattern reinforces the interpretation that the Q4 spatial divergence is a market-driven phenomenon. Private and foreign firms respond to market incentives: better export access combined with the productive environment of large clusters. SOEs, operating under different objectives and subject to government direction, do not reallocate toward Q4 clusters in response to the same market signal.



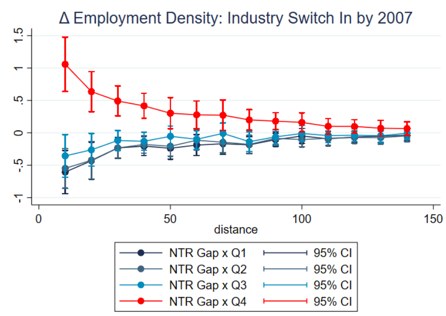
(a) Incumbents



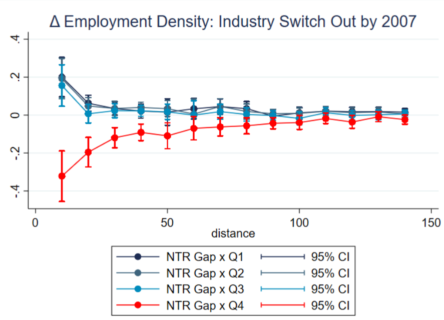
(b) Exits



(c) Entry

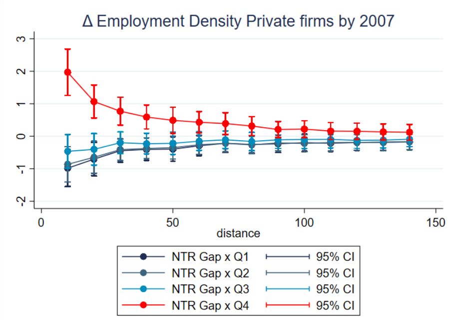


(d) Industry In-Switching

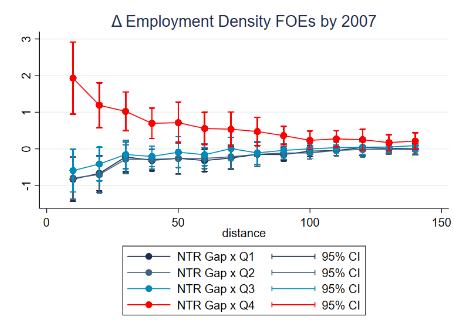


(e) Industry Out-Switching

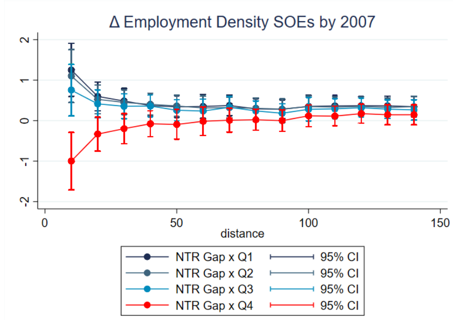
Figure 6: Decomposed cumulative change in employment density by 2007, by firm margin and ring. Each panel plots $\hat{\beta}_{q,2007}$ from equation 2 estimated on the margin-specific component of density change. Incumbents are firms present in the same industry-prefecture in both 1998 and 2007; exits are firms that departed the sample; entry is firms appearing for the first time; in-switching (out-switching) are firms joining (leaving) the industry from (to) another three-digit industry. Standard errors clustered at the three-digit industry level; 95% confidence intervals shown.



(a) Private



(b) Foreign-Owned



(c) State- and Collectively Owned

Figure 7: Decomposed cumulative change in employment density by 2007, by ownership type and ring. Each panel plots $\hat{\beta}_{q,2007}$ from equation 2 estimated on the ownership-specific component of density change. Private firms are domestic private enterprises; foreign-owned includes joint ventures and wholly foreign-owned enterprises; state- and collectively owned includes state-owned enterprises and collective enterprises. Standard errors clustered at the three-digit industry level; 95% confidence intervals shown.

5. Robustness

5.1. Infrastructure Controls

A first concern is that road expansion and high-speed rail connections were expanding rapidly over our sample period and could independently drive spatial concentration. Local governments may respond to trade-driven growth by investing in transport infrastructure, which would further reduce internal trade costs and amplify agglomeration. If such investment correlates with the NTR gap across industries, the estimates in equation 2 would overstate the direct effect of trade uncertainty reduction.

To address this concern, we augment equation 2 with the change in total road length within each prefecture relative to 1998 and a binary indicator for the establishment of a high-speed railway connection. Figure 8 shows the cumulative 2007 distance profile with these additional controls. The Q4 pattern is unchanged, the magnitude and attenuation are all robust to the inclusion of infrastructure controls.

5.2. Prefecture-Level Trade Exposure

A second concern is that prefectures differ systematically in their aggregate NTR exposure, reflecting differences in their industry mix. Large prefectures are more likely to host Q4 clusters and may also be more heavily exposed to the aggregate trade shock, so changes in Q4 density could reflect a prefecture-level growth story rather than industry-specific liberalization.

We address this by including the employment-weighted average NTR gap at the prefecture level as an additional control variable. This captures the direct effect of a prefecture's overall NTR gap exposure and isolates the industry-specific component of the NTR gap interaction. The results, also shown in Figure 8, remain essentially unchanged.

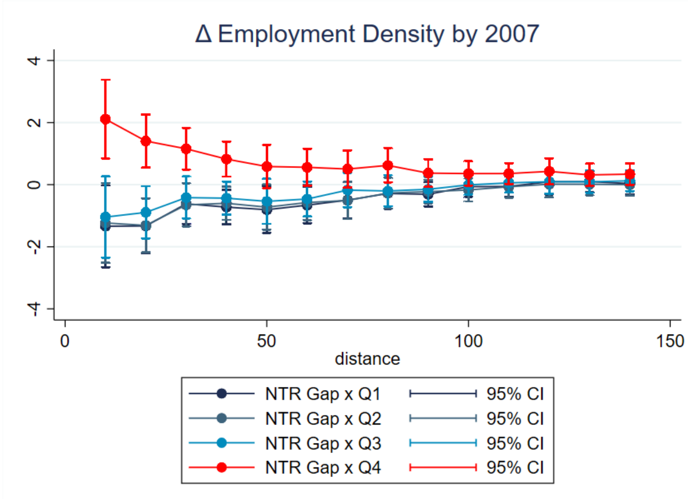


Figure 8: Cumulative change in employment density by 2007, with additional controls. Plotted coefficients are $\hat{\beta}_{q,2007}$ from equation 2 extended to include the change in prefecture road length, a high-speed railway indicator, and the employment-weighted average prefecture NTR gap. Standard errors clustered at the three-digit industry level; 95% confidence intervals shown. The 0–10km ring is excluded.

5.3. Pooled Specification With Prefecture-By-Year Fixed Effects

A third concern is that Q4 clusters are concentrated in larger, faster-growing prefectures, so their differential growth could reflect prefecture-specific time trends rather than a response to the industry-level NTR gap. Equation 3 in Section 3 addresses this directly. By pooling across years and including prefecture-by-year fixed effects μ_{pt}, the specification absorbs all prefecture-specific trends, including the effects of infrastructure investment and aggregate trade exposure documented in the two preceding subsections.

Table 3 reports results pooled across rings and post-2001 years. Column (1) uses the level change in employment density; column (2) uses the inverse-hyperbolic sine (ihs) transformation, which captures relative changes while remaining defined at zero. In both specifications, Q4 clusters grow more than Q1–Q3 clusters in response to a larger NTR gap following WTO accession. The Q4 coefficient is 0.715 (standard error 0.242) in levels and 0.178 (0.058) in the ihs specification, both significant at the 1% level. The ihs result is particularly informative: even in relative terms, initially large clusters expand the most, ruling out the mechanical interpretation that absolute density increases appear larger in Q4 simply because Q4 baseline density is higher.

Table 3: Pooled Regression

	(1) Change in density	(2) Growth in density (ihs)
NTRgap × Q1 × WTO	0.151	0.0777**
	(0.106)	(0.0374)
NTRgap × Q2 × WTO	0.170*	0.0807**
	(0.103)	(0.0376)
NTRgap × Q3 × WTO	0.250**	0.104**
	(0.121)	(0.0427)
NTRgap × Q4 × WTO	0.715***	0.178***
	(0.242)	(0.0578)
City x Year FE	Yes	Yes
Industry FE	Yes	Yes
N	2459786	2459786

Column 1 shows total changes in density relative to 1998. Density is measured in workers per 10k square meters. Column 2 shows changes in inverse-hyperbolic sine transformed density (×100). Standard errors (in parentheses) are clustered at the 3-digit industry level. For consistency with the previous figures the 0-10km ring is excluded.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6. Co-Agglomeration

The Matthew effect documented in Section 4 operates through a single industry: export-exposed Q4 clusters grow while smaller clusters do not. But this direct effect understates the full reinforcement of initial advantage. A cluster's dominance also deepens through co-agglomeration: as a large cluster expands in response to trade liberalization, the suppliers and customers that depend on it have an incentive to locate nearby, further entrenching the cluster's competitive position. Ellison et al. (2010) establish that IO linkages are the primary driver of industrial co-location in the US. Helm (2020) finds that trade shocks propagate locally to other industries, with the channel depending on whether linkages are through production or labor. We ask whether the same reinforcing dynamic is present in

China's trade liberalization episode.

To measure co-agglomeration, we modify equation 2 by replacing the left-hand side with the weighted average density change of industries related to j. The estimating equation becomes

$$\sum_{k\neq j}\omega_{jk}\Big(Y^{rt}_{pk}-Y^{r,1998}_{pk}\Big)=\sum_{q=1}^{4}\beta^{rt}_{q}\,\mathrm{NTR}_j\times Q_{pjq}+X^{rt}_{pj}+\mu_P+\gamma_s+\varepsilon^{rt}_{pj} \tag{4}$$

where the left-hand side is the weighted change in density of industries related to j, with weights ω_{jk} capturing the strength of the linkage between industry j and each related industry k. The right-hand side is identical to equation 2: the coefficient β^{rt}_q identifies the effect of industry j's reduction in trade policy uncertainty on the density of its related industries, separately by initial cluster size quartile q, ring r, and year t.

We construct three linkage measures, each corresponding to a distinct economic channel.

Input linkages. We use the 2002 Chinese input-output tables to measure upstream linkages. For each industry pair (j,k), ω_{jk} equals the cost expenditure share of industry j spent on inputs from industry k. When a Q4 cluster expands, it demands more locally sourced inputs, attracting suppliers to co-locate. Having a deep pool of nearby suppliers then lowers input costs and reduces the frictions of sourcing, reinforcing the cluster's advantage.

Output linkages. Using the same IO tables, we measure downstream linkages. Here ω_{jk} equals the share of industry j's output purchased by industry k. As a Q4 cluster grows in output, downstream buyers benefit from closer proximity to a reliable local supplier base, giving them an incentive to locate nearby as well.

Labor pooling. We use the 2000 Chinese Population Census, which reports employment by detailed industry and occupation, to measure the extent to which two industries draw from the same occupational pool. For each industry pair (j,k), ω_{jk} equals the correlation of their occupation employment shares. Industries with highly correlated occupation profiles compete for the same workers. Negative correlations are set to zero to avoid negative employment weights. Unlike supply-chain linkages, labor pooling generates competition rather than complementarity: when a Q4 cluster expands and bids up local wages, neighboring industries that draw from the same labor pool face higher costs and are crowded out.

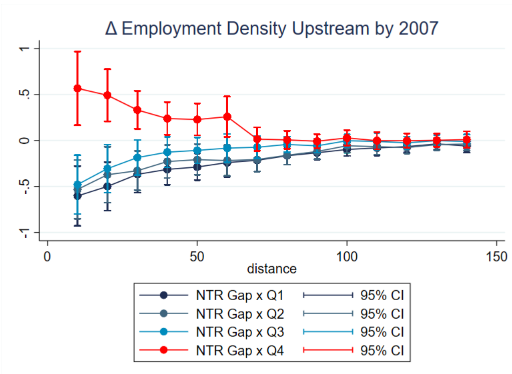
6.1. Results

Figure 9 shows the cumulative 2007 co-agglomeration effects for all three linkage types, plotting $\hat{\beta}_{q,2007}$ from equation 4 across the distance profile by initial size quartile.

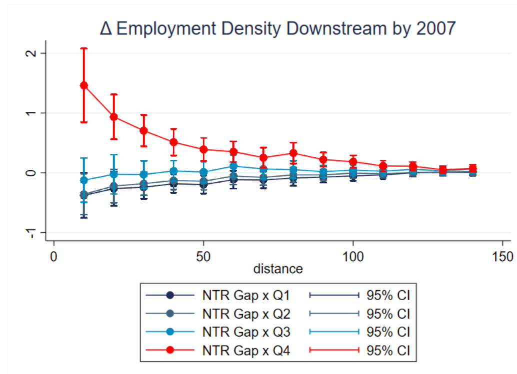
Panels (a) and (b) confirm that co-agglomeration through supply-chain linkages is a Q4 phenomenon. Upstream input suppliers and downstream output buyers both grow in employment density around clusters of the trade-exposed industry, but only when that cluster was already large in 1998. The spatial profile mirrors the main results exactly: effects peak at 10–20km from the cluster center and attenuate with distance. This pattern is consistent with Ellison et al. (2010), who find that IO linkages drive industry co-location. For Q4 clusters, co-agglomeration is a further source of advantage. As the cluster grows, it pulls in the suppliers and customers that further reduce its costs and deepen its local ecosystem, compounding the initial impetus from trade liberalization.

Panel (c) shows the opposite pattern for labor-pooling industries: negative and significant density effects, again concentrated in Q4. Industries that compete for the same workers lose employment density as a neighboring export cluster expands. The entry of new private and foreign firms documented in Section 4.3 raises local labor demand for a specific pool of workers. Industries sharing the same occupational pool face higher labor costs, deterring further entry and eroding their local presence. The crowding-out effect is local in scope, consistent with imperfect labor mobility across regions.

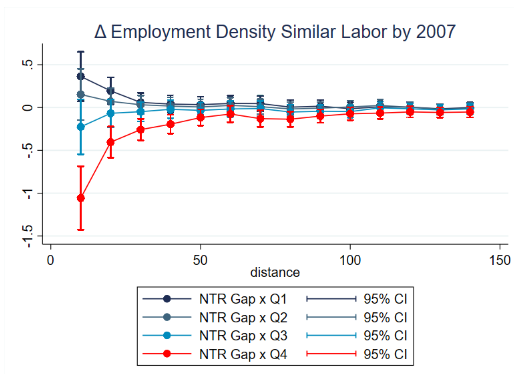
Together, these results reveal co-agglomeration as a second dimension of the Matthew effect. Q4 clusters do not just grow within their own industry. Initial advantage in industry j begets further advantage for j through the co-location of j's upstream and downstream partners, even as it imposes a localized cost on industries sharing j's labor pool.



(a) Upstream linkages



(b) Downstream linkages



(c) Labor pooling

Figure 9: Cumulative co-agglomeration effects by 2007. Each panel plots $\hat{\beta}_{q,2007}$ from equation 4, where the dependent variable is the weighted average density change of industries related to the trade-exposed industry j. Panel (a): upstream input suppliers, with weights equal to the cost expenditure share of j spent on inputs from industry k (2002 Chinese IO tables). Panel (b): downstream output buyers, with weights equal to the share of j's output purchased by k (same source). Panel (c): labor-pooling competitors, with weights equal to the cross-industry correlation of occupation employment shares (2000 Chinese Population Census). Standard errors clustered at the three-digit industry level; 95% confidence intervals shown. The 0–10km ring is excluded.

7. Theoretical Framework

We develop a spatial Melitz model to rationalize the empirical patterns documented in Sections 4–6. The model incorporates two features of China's development context: manufacturers sell exclusively to foreign markets, isolating the export access channel; and rural workers migrate to cities in response to wage differentials, providing elastic labor supply to expanding clusters. Two predictions emerge. Cities whose firm productivity distribution is shifted further to the right attract more workers and command higher wages. A reduction in export tariffs amplifies the wage gap between larger and smaller cities, causing the former to grow disproportionately. The model provides a micro-founded account of the Matthew effect we document empirically.

Throughout this section, we index cities by $k \in \{1, 2\}$ to distinguish from the industry index j used in the empirical sections.

7.1. Environment

Spatial structure. The economy consists of a rural area, two cities $k \in \{1, 2\}$, and a foreign country. Firms locate in cities and produce exclusively for export. The iceberg trade cost $\tau \in (0, 1)$ represents the fraction of output lost in transit: delivering one unit abroad requires shipping $1/(1 - \tau)$ units. We model China's WTO accession in 2001 as an unexpected, permanent reduction from τ^h to $\tau^l < \tau^h$.

Three periods. Period 0: all workers reside in the rural area; no manufacturing exists. Period 1 (corresponding to the pre-WTO period around 1998): firms enter under tariff τ^h and rural workers migrate to cities. Period 2 (post-2001 WTO accession): the tariff falls unexpectedly to τ^l, new firms enter, and rural workers who did not migrate in period 1 reallocate across destinations. Workers already in a city cannot switch cities. Period 1 thus establishes the initial spatial distribution; period 2 studies how trade liberalization reshapes it.

Labor supply. Workers choose between the rural area (earning w_R from home production) and city k (earning w_k). Idiosyncratic location preferences are drawn from a Gumbel distribution with scale parameter $\alpha > 0$. The probability that a rural worker migrates to destination $k \in \{R, 1, 2\}$ follows the logit formula:

$$\lambda_k(\mathbf{w}) = \frac{e^{\alpha w_k}}{\sum\limits_{m \in \{R,1,2\}} e^{\alpha w_m}}, \quad k \in \{R, 1, 2\} \tag{5}$$

where $\mathbf{w} = (w_R, w_1, w_2)$. Normalizing total labor to 1, city k's employment in period 1 is $L_{k1} = \lambda_k(\mathbf{w}_1)$. In period 2, the remaining rural pool $L_{R1} = 1 - L_{11} - L_{21}$ faces the same logit choice under period-2 wages, while period-1 city residents remain in place. City k's labor supply in period 2 is:

$$L_k^s(\mathbf{w}_2) = L_{k1} + L_{R1}\,\lambda_k(\mathbf{w}_2), \quad k \in \{1, 2\} \tag{6}$$

Foreign demand. Foreign consumers have CES preferences over Chinese varieties with elasticity of substitution $\sigma > 1$. China is a small open economy: the foreign price index $\mathbb{P}$ and total foreign expenditure on Chinese goods $\mathbb{W}$ are exogenous. Demand for variety i at price p_i is $x_i = (p_i/\mathbb{P})^{-\sigma}(\mathbb{W}/\mathbb{P})$.

Firms. Firms use labor as the sole input. Producing q units requires a fixed cost $f > 0$ (in labor units) and a variable input inversely proportional to productivity ϕ:

$$\ell(\phi) = f + \frac{q}{\phi} \tag{7}$$

Under CES monopolistic competition, a firm with productivity ϕ in city k sets the export price at a constant markup over the delivered marginal cost:

$$p_k(\phi) = \frac{1}{1-\tau} \cdot \frac{\sigma}{\sigma - 1} \cdot \frac{w_k}{\phi} \tag{8}$$

Substituting into CES demand yields firm revenue and profit:

$$r_k(\phi) = \mathbb{W}\left(\frac{\mathbb{P}\,\rho\,(1-\tau)\,\phi}{w_k}\right)^{\sigma-1}, \qquad \rho \equiv \frac{\sigma-1}{\sigma}$$
$$\pi_k(\phi) = \frac{r_k(\phi)}{\sigma} - f w_k$$

7.2. Entry, Selection, And Free Entry

Entry. Each city has an unbounded pool of prospective entrants. Entry requires a sunk cost of $F > 0$ labor units (monetary cost Fw_k). Upon entering, a firm draws productivity ϕ from a city-specific distribution $G_k(\phi)$ with density $g_k(\phi)$ and support $[\ell_k, \infty)$. Active firms face a constant per-period exit probability $\delta \in (0,1)$.

Selection. A city-specific cutoff ϕ_k^* separates profitable producers ($\phi \geq \phi_k^*$) from firms that exit immediately. Setting $\pi_k(\phi_k^*) = 0$ implies the cutoff firm earns revenue exactly $\sigma f w_k$. Following Melitz (2003), the average observed productivity $\tilde{\phi}_k$ is the level generating the same revenue as the average active firm:

$$\tilde{\phi}_k \equiv \left[\frac{1}{1-G_k(\phi_k^*)}\int_{\phi_k^*}^{\infty} \phi^{\sigma-1}\, g_k(\phi)\,\mathrm{d}\phi\right]^{1/(\sigma-1)} \tag{9}$$

Average profit among active firms is:

$$\bar{\pi}_k = \left[\left(\frac{\tilde{\phi}_k}{\phi_k^*}\right)^{\sigma-1} - 1\right] f w_k \tag{10}$$

Free entry. In steady state, δM_k firms exit per period. Replacing them requires $\delta M_k/[1 - G_k(\phi_k^*)]$ entry attempts. The free entry condition equates expected profits to the sunk cost:

$$\left[1 - G_k(\phi_k^*)\right]\frac{\bar{\pi}_k}{\delta} = F w_k \tag{11}$$

Substituting equation (10) into equation (11), the wage cancels (since both entry cost F and fixed production cost f are in labor units), yielding[5]:

$$\frac{\delta F}{1-G_k(\phi_k^*)} = \left[\left(\frac{\tilde{\phi}_k(\phi_k^*)}{\phi_k^*}\right)^{\sigma-1} - 1\right] f \tag{12}$$

7.3. General Equilibrium

Goods market clearing. Total foreign expenditure on Chinese goods equals aggregate revenue:

$$\sum_k M_k \bar{r}_k = \mathbb{W} \tag{13}$$

Labor market clearing. Entry labor demand in steady state is:

$$L_k^e = F\,\frac{\delta M_k}{1-G_k(\phi_k^*)} \tag{14}$$

Production labor demand, aggregated over active firms, is:

$$L_k^p = M_k\left[f + \frac{\bar{r}_k\,(\sigma-1)}{\sigma w_k}\right] \tag{15}$$

Combining entry and production labor demands with the free entry condition (11), all fixed costs cancel (see Appendix A.3) and labor market clearing reduces to:

$$\frac{M_k \bar{r}_k}{w_k} = L_k^s(\mathbf{w}) \tag{16}$$

Since labor is the sole input and all costs are paid in wages, total revenue equals the total wage bill.

Equilibrium definition. A Spatial-Melitz Equilibrium is a set $\{w_k, M_k, \phi_k^*\}_{k=1}^2$ such that, for each city k: (i) the cutoff ϕ_k^* satisfies (12); (ii) the firm mass M_k satisfies (13);

[5] The derivation is in Appendix A.1.

(iii) the wage w_k satisfies (16).

7.4. Predictions

To obtain closed-form results, we specialize the productivity distribution. Following Helpman et al. (2004) and Bakker (2021), we adopt a Pareto specification with a city-specific location parameter.

Assumption 1. *The city-specific productivity distribution has CDF $G_k(\phi) = 1 - (\ell_k/\phi)^\kappa$ for $\phi \geq \ell_k > 0$, with common shape parameter $\kappa > \sigma - 1$.*

Cities differ solely in the location parameter ℓ_k. A higher ℓ_k shifts the entire distribution to the right: for any fixed threshold $x \geq \ell_k$, the survival probability $1 - G_k(x) = (\ell_k/x)^\kappa$ is increasing in ℓ_k, so a city with a higher location parameter first-order stochastically dominates one with a lower location parameter. The implied productivity distribution therefore places more mass above any given absolute level. Note that the tail decay rate is governed by κ, which is common across cities; what differs is the position of the distribution, not the shape of its tail. The restriction $\kappa > \sigma - 1$ ensures convergence of the integral in (9).

Under Assumption 1, the free-entry condition (12) admits a closed-form solution.

Proposition 1. *Under Assumption 1, the equilibrium productivity cutoff and average observed productivity in city k are:*

$$\phi_k^* = \ell_k \left[\frac{(\sigma - 1) f}{(\kappa + 1 - \sigma) \delta F} \right]^{1/\kappa} \tag{17}$$

$$\tilde{\phi}_k = \left[\frac{\kappa}{\kappa + 1 - \sigma} \right]^{1/(\sigma-1)} \phi_k^* \tag{18}$$

Both are proportional to ℓ_k and independent of w_k, τ, $\mathbb{P}$, and M_k.

The proof is in Appendix A.2. The key property is that the ranking of cities by cutoff and average productivity depends only on ℓ_k and is invariant to wages and trade policy. The ratio $\tilde{\phi}_k/\phi_k^*$ is a constant, a standard implication of the Pareto distribution (Melitz 2003), so the cutoff and the average always move in proportion.

Under Assumption 1, substituting the Pareto survival probability and the constant ratio $\tilde{\phi}_k/\phi_k^*$ into (16) yields:

$$\bar{r}_k = \frac{\sigma f \kappa}{\kappa + 1 - \sigma}\, w_k, \qquad M_k = \frac{\kappa + 1 - \sigma}{\sigma f \kappa}\, L_k^s(\mathbf{w})$$

Average revenue is proportional to w_k and independent of ℓ_k and τ. The firm mass is proportional to city employment and independent of w_k.

Wage determination. The tariff enters equilibrium through the zero-profit condition, which links revenue to the cutoff. Substituting $\phi_k^* = \ell_k \Lambda$, where $\Lambda \equiv [(\sigma - 1)f/((\kappa + 1 - \sigma)\delta F)]^{1/\kappa}$ collects structural parameters, the zero-profit condition pins down the wage:

$$w_k = \left[\frac{\mathbb{W}\Big(\mathbb{P}\,\rho\,(1-\tau)\,\Lambda\Big)^{\sigma-1}}{\sigma f}\right]^{1/\sigma} \ell_k^{(\sigma-1)/\sigma} \tag{19}$$

The wage is increasing in both ℓ_k and $(1 - \tau)$: cities with a rightward-shifted productivity distribution pay higher wages, and a tariff reduction raises wages everywhere.

Cross-city wage ratio. Taking the ratio of (19) across cities, all common factors cancel:

$$\frac{w_1}{w_2} = \left(\frac{\ell_1}{\ell_2}\right)^{(\sigma-1)/\sigma} \tag{20}$$

The cross-city wage ratio depends only on the ratio of location parameters and is *independent of the tariff*. Since $(\sigma - 1)/\sigma \in (0, 1)$, a city with a higher ℓ_k commands a strict but less-than-proportional wage premium.

Proposition 2. *Under Assumption 1, the equilibrium wage w_k is strictly increasing in ℓ_k.*

The proof is in Appendix A.4. The mechanism is direct: a higher location parameter raises the productivity cutoff and average productivity (Proposition 1), generating higher average revenues that bid up the local wage. This rationalizes the empirical finding that larger, more productive clusters command wage premia.

The second assumption links the location parameter to initial city size.

Assumption 2. *The city-specific productivity location parameter ℓ_k is strictly increasing in initial city employment e_k.*

Assumption 2 reflects the observation that historically larger cities accumulated more productive entrepreneurs through sorting, learning-by-doing, and agglomeration spillovers during earlier reform periods. In the empirical setting, city 1 (with $e_1 > e_2$) maps to the Q4 quartile of initial cluster size.

Under Assumptions 1 and 2, the model delivers its central prediction.

Proposition 3. *Under Assumptions 1 and 2, when $e_1 > e_2$: (i) $L_1 > L_2$ in equilibrium; and (ii) a reduction in τ increases L_1/L_2.*

The mechanism proceeds as follows. Since $e_1 > e_2$, Assumption 2 gives $\ell_1 > \ell_2$, and Proposition 2 gives $w_1 > w_2$. The logit migration model then implies $L_{11} > L_{21}$ in period 1. In period 2, both the inherited stock and incremental migration favor city 1, establishing part (i). For part (ii), write $w_1 = Rw_2$ where $R = (\ell_1/\ell_2)^{(\sigma-1)/\sigma} > 1$ is constant in τ. The absolute wage gap is $(R-1)w_2$. From (19), $w_2 \propto (1-\tau)^{(\sigma-1)/\sigma}$: when τ falls, w_2 rises and the absolute gap widens. The logit migration ratio $\lambda_1(\mathbf{w}_2)/\lambda_2(\mathbf{w}_2) = e^{\alpha(w_1-w_2)}$ is strictly increasing in the absolute gap, so a wider gap redirects more of the remaining rural pool toward city 1, raising L_1^s/L_2^s and, by (16), the firm mass ratio M_1/M_2. The full proof is in Appendix A.5.

This divergence is entirely *entry-driven*: the free entry margin is the operative spatial adjustment channel. Incumbents with productivity draws fixed before the tariff change face unchanged exit probability δ. In the Pareto equilibrium, individual firm employment depends only on the ratio ϕ/ℓ_k, which is fixed; incumbent employment therefore does not expand when τ falls. This aligns with the decomposition in Section 4.3, where new firm entry accounts for virtually all Q4 density growth while incumbents show no expansion. The model's entrants are profit-maximizing agents responding to market incentives, the natural counterpart of the private and foreign firms that drive all Q4 growth in Section 4.4. State-owned enterprises, operating under non-market objectives, lie outside the model's scope.

The model is silent on two further empirical findings. The distance gradient (effects peaking at 10–20km and decaying by 100km) requires within-city spatial structure (land markets, commuting costs) that the current framework abstracts from. The co-agglomeration patterns in Section 6 require multi-industry input-output linkages and cross-industry labor market competition, neither of which the single-industry model incorporates. Both are natural extensions.

8. Conclusion

This paper shows that China's WTO accession caused spatial divergence in manufacturing: industries experiencing larger reductions in trade policy uncertainty saw their pre-existing largest clusters grow in employment density, while smaller clusters did not benefit. Using geolocated firm-level data from the Annual Survey of Industrial Firms and the NTR gap of Pierce and Schott (2016) as our identifying variation, we find that only the initially largest quartile of industry-prefecture clusters expanded after 2001. The effects peak at 10–20km from the cluster center, attenuate with distance, and become statistically insignificant beyond 100km. For the average cluster in the top quartile, a one-standard-deviation larger NTR gap reduction implies a 21.2% increase in employment density at 10–20km by 2007. Decomposing these changes, we find that growth is driven entirely by new firm entry, particularly by private and foreign firms; incumbent firms do not expand, and may exit.

Export-induced agglomeration also has consequences for related industries. In 6, we show that positive density spillovers propagate to upstream and downstream industries in large clusters. Industries that compete for the same workers, by contrast, lose density when a neighboring export-intensive industry expands, pointing to a wage competition channel and imperfect inter-industry labor mobility. The Spatial-Melitz framework in 7 rationalizes all of these findings: when cities differ in their productivity distributions and larger cities have thicker upper tails, a tariff reduction amplifies the initial advantage of the largest agglomeration.

Several limitations bound the scope of these conclusions. The model abstracts from between-city labor mobility and import competition, both of which could attenuate or redirect the spatial effects we document. Our spatial measure, employment density in concentric rings, also cannot separately identify within-city location choices from between-city reallocation.

These findings carry a clear implication for policy: trade openness tends to reinforce rather than diffuse spatial inequality, concentrating gains in already-dominant industrial clusters. The labor market crowding result suggests that policies facilitating inter-industry labor reallocation, such as Hukou reform, could broaden the geographic incidence of trade liberalization gains.

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Appendix A. Proofs And Derivations

A.1. Derivation Of The Combined Free-Entry Condition

The free entry condition (11) states $[1 - G_k(\phi_k^*)]\, \bar{\pi}_k/\delta = F w_k$. Substituting average profit from equation (10):

$$\Big[1 - G_k(\phi_k^*)\Big] \frac{1}{\delta} \left[\left(\frac{\tilde{\phi}_k}{\phi_k^*} \right)^{\sigma-1} - 1 \right] f w_k = F w_k$$

Since both entry cost F and fixed production cost f are denominated in labor units, the wage $w_k > 0$ appears on both sides and cancels, yielding equation (12):

$$\frac{\delta F}{1 - G_k(\phi_k^*)} = \left[\left(\frac{\tilde{\phi}_k(\phi_k^*)}{\phi_k^*} \right)^{\sigma-1} - 1 \right] f$$

The left side is increasing in ϕ_k^* (since $1 - G_k(\phi_k^*)$ is decreasing); the right side depends on ϕ_k^* only through $\tilde{\phi}_k(\phi_k^*)$, which is increasing. Under standard regularity conditions on G_k, there exists a unique solution ϕ_k^* that depends only on G_k and (f, F, δ, σ).

A.2. Proof Of Proposition 1

Under Assumption 1, the survival probability is $1 - G_k(\phi) = (\ell_k/\phi)^\kappa$.

Step 1: Average productivity. Substituting the Pareto density into equation (9):

$$\tilde{\phi}_k^{\sigma-1} = \left(\frac{\phi_k^*}{\ell_k} \right)^\kappa \kappa\, \ell_k^\kappa \int_{\phi_k^*}^{\infty} \phi^{\sigma-2-\kappa}\, \mathrm{d}\phi$$

Since $\kappa > \sigma - 1$, the exponent $\sigma - 2 - \kappa < -1$ and the integral converges to $(\phi_k^*)^{\sigma-1-\kappa}/(\kappa + 1 - \sigma)$. Substituting:

$$\tilde{\phi}_k^{\sigma-1} = \frac{\kappa}{\kappa + 1 - \sigma} (\phi_k^*)^{\sigma-1}$$

Taking the $(\sigma - 1)$-th root confirms equation (18).

Step 2: Cutoff productivity. From Step 1, $(\tilde{\phi}_k/\phi_k^*)^{\sigma-1} = \kappa/(\kappa + 1 - \sigma)$. Substituting into equation (12) with survival probability $(\ell_k/\phi_k^*)^\kappa$:

$$\delta F \left(\frac{\phi_k^*}{\ell_k} \right)^\kappa = \left[\frac{\kappa}{\kappa + 1 - \sigma} - 1 \right] f = \frac{(\sigma - 1)\, f}{\kappa + 1 - \sigma}$$

Solving for ϕ_k^* confirms equation (17). Both ϕ_k^* and $\tilde{\phi}_k$ are proportional to ℓ_k and independent of w_k, τ, $\mathbb{P}$, and M_k.

A.3. Derivation Of The Simplified Labor Market Clearing Condition

Total labor demand is $L_k^e + L_k^p$. Using the free entry condition (11), entry labor demand satisfies $L_k^e = F\,\delta M_k/[1 - G_k(\phi_k^*)] = M_k\bar{\pi}_k/w_k$. Adding production labor demand from equation (15) and substituting $\bar{\pi}_k = \bar{r}_k/\sigma - f w_k$ from equation (10):

$$
\begin{aligned}
L_k^e + L_k^p &= \frac{M_k\bar{\pi}_k}{w_k} + M_k\left[f + \frac{\bar{r}_k\,(\sigma-1)}{\sigma w_k}\right] \\
&= \frac{M_k}{w_k}\left(\frac{\bar{r}_k}{\sigma} - f w_k\right) + M_k f + \frac{M_k\bar{r}_k(\sigma-1)}{\sigma w_k} \\
&= \frac{M_k\bar{r}_k}{\sigma w_k} - M_k f + M_k f + \frac{M_k\bar{r}_k(\sigma-1)}{\sigma w_k} = \frac{M_k\bar{r}_k}{w_k}
\end{aligned}
$$

Setting this equal to labor supply gives equation (16). The result states that total revenue in city k equals the total wage bill, which holds because labor is the only production factor and all costs (fixed, variable, and entry) are paid in wages.

A.4. Proof Of Proposition 2

From equation (20), $w_1/w_2 = (\ell_1/\ell_2)^{(\sigma-1)/\sigma}$. Since $(\sigma-1)/\sigma > 0$, this ratio is strictly increasing in ℓ_1 (holding ℓ_2 fixed) and strictly decreasing in ℓ_2 (holding ℓ_1 fixed). For the absolute wage level, equation (19) shows $w_k \propto \ell_k^{(\sigma-1)/\sigma}$, which is strictly increasing in ℓ_k for fixed $(\mathbb{W}, \mathbb{P}, \tau)$. Under the small open economy assumption, $\mathbb{P}$ is exogenous, so this comparative static is well-defined. In the two-city case, the aggregate income identity $\sum_k w_k L_k^s(\mathbf{w}) = \mathbb{W}$ pins down the wage level given the ratio from equation (20); the logit labor supply is increasing in wages, ensuring a unique positive solution.

A.5. Proof Of Proposition 3

Part (i). By Assumption 2, $e_1 > e_2$ implies $\ell_1 > \ell_2$. By Proposition 2, $w_1 > w_2$. Since $\lambda_k(\mathbf{w})$ in equation (5) is strictly increasing in w_k (holding other wages fixed), $\lambda_1(\mathbf{w}_1) > \lambda_2(\mathbf{w}_1)$, so $L_{11} > L_{21}$. In period 2, $L_k^s(\mathbf{w}_2) = L_{k1} + L_{R1}\lambda_k(\mathbf{w}_2)$. Since $w_1 > w_2$ persists (the wage ratio (20) is invariant to τ), both the inherited stock and incremental migration favor city 1. Hence $L_1^s(\mathbf{w}_2) > L_2^s(\mathbf{w}_2)$.

Part (ii). Write $w_1 = R w_2$ where $R = (\ell_1/\ell_2)^{(\sigma-1)/\sigma} > 1$ is constant in τ. The absolute wage gap is $w_1 - w_2 = (R-1)w_2$. From equation (19), $w_2 \propto (1-\tau)^{(\sigma-1)/\sigma}$: when τ falls,

$(1-\tau)$ rises, w_2 increases, and the absolute gap $(R-1)w_2$ widens. The period-2 logit migration ratio for rural workers is:

$$\frac{\lambda_1(\mathbf{w}_2)}{\lambda_2(\mathbf{w}_2)} = e^{\alpha(w_1-w_2)}$$

Since $w_1 - w_2$ has increased, this ratio rises: a larger share of the remaining rural pool L_{R1} migrates to city 1. In period 2:

$$\frac{L_1^s}{L_2^s} = \frac{L_{11} + L_{R1}\,\lambda_1(\mathbf{w}_2)}{L_{21} + L_{R1}\,\lambda_2(\mathbf{w}_2)}$$

The numerator terms $L_{11} > L_{21}$ are fixed. The migration terms $L_{R1}\lambda_k(\mathbf{w}_2)$ shift toward city 1. Both effects raise L_1^s/L_2^s. By equation (16), $M_k \propto L_k^s$, so M_1/M_2 also rises.

Appendix B. Additional Data Tables

Table A1: Summary of ASIF Firm Data by Year

Year	Number of Firms	Value Added (trillion RMB)	Output (trillion RMB)	Employment (million)	Exports (trillion RMB)	Net Fixed Assets (trillion RMB)
1998	165,110	1.94	6.77	61.96	1.08	4.41
1999	162,029	2.16	7.27	58.05	1.15	4.73
2000	162,884	2.54	8.57	55.59	1.46	5.18
2001	171,243	2.83	9.54	54.41	1.62	5.54
2002	181,552	3.30	11.08	55.21	2.01	5.95
2003	196,213	4.20	14.23	57.48	2.69	6.61
2004	278,984	5.72	20.17	66.16	4.05	7.95
2005	271,830	7.22	25.16	68.96	4.77	8.95
2006	301,957	9.11	31.65	73.58	6.05	10.57
2007	336,763	11.70	40.50	78.74	7.33	12.34

Note: This table reports aggregate statistics from the Annual Survey of Industrial Firms (ASIF) by year. The sample includes all industrial firms in the database, including non-manufacturing industrial firms, before applying the sample restrictions described in Section 2. Employment is in millions of workers. Value added, output, exports, and net value of fixed assets are in trillions of nominal RMB.

Table A2: Geocoding Precision

Geocoding confidence	Frequency	Share (%)
Error $\leq$ 100m at 91% confidence	1,108,794	48.86
Error $\leq$ 100m at 89% confidence	49,109	2.21
Error $\leq$ 100m at 88% confidence	11,720	0.53
Error $\leq$ 100m at 88–79% confidence	1,077,743	48.40

Note: This table reports the distribution of geocoding confidence scores for all firm-year observations geocoded via Baidu Map API v3.0. Each row corresponds to a confidence tier: the first three rows represent cases where the API assigns at least a 88% probability that the geocoding error is within 100 meters; the final row covers cases where confidence ranges between 79% and 88%. Observations that could not be resolved to geographic coordinates are excluded from the analytical sample.